# ***"UniMate University"***

## *Requirement Analysis - MVP Project*

## MERN Stack MVP Web Application

### 1. Functional Requirements:

**a. Authentication:**

**i. Student:**

1. Can login and logout securely.
2. Can update their password.

**ii. Faculty:**

1. Can login and logout securely.
2. Can update their password.

**iii. Admin:**

1. Can login and logout securely.
2. Can update their password.

**b. Profile Management:**

**i. Student:**

1. Can manage and update their profile information.
2. Can update certain fields only.

**ii. Faculty:**

1. Can manage and update their profile information.
2. Can update certain fields only.

**iii. Admin:**

1. Can manage and update their profile information.
2. Can update certain fields only.

**c. Academic Procedures:**

**i. Student:**

1. Can enroll in offered courses for a specific semester.
2. Can view their class schedule.
3. Can view their grades and results.
4. Can view notice boards and events.

**ii. Faculty:**

1. Can manage students' grades.
2. Can access students' personal and academic information.

**iii. Admin:**

1. **Can manage multiple processes:**
   a. Semesters;
   b. Courses;
   c. Offered Courses;
   d. Sections;
   e. Rooms;
   f. Buildings;

**d. User Management:**

**i. Admin:**

1. Can manage multiple types of users.
2. Can block/unblock users.
3. Can change users' passwords.

## 2. Data Model:

<table>
<tr><th colspan="2">Academic Faculty</th><th colspan="2">Academic Department</th></tr>
<tr><td colspan="2">• _id<br>• name<br>• createdAt<br>• updatedAt</td><td colspan="2">• _id<br>• name<br>• academicFaculty<br>• createdAt<br>• updatedAt</td></tr>
<tr><th colspan="2">Management Department</th><th colspan="2">Course</th></tr>
<tr><td colspan="2">• _id<br>• name<br>• createdAt<br>• updatedAt</td><td colspan="2">• _id<br>• title<br>• prefix<br>• code<br>• credits<br>• prerequisiteCourse:<br>  ○ course<br>  ○ isDeleted<br>• isDeleted<br>• createdAt<br>• updatedAt</td></tr>
<tr><th colspan="2">Academic Semester</th><th colspan="2">Semester Registration</th></tr>
<tr><td colspan="2" rowspan="2">• _id<br>• name<br>• year<br>• code<br>• startMonth<br>• endMonth<br>• createdAt<br>• updatedAt</td><th>Client Model</th><th>Database Model</th></tr>
<tr><td>• academicSemester<br>• startDay<br>• endDay<br>• startTime<br>• endTime<br>• minCredit<br>• maxCredit</td><td>• _id<br>• academicSemester<br>• status<br>• startDate<br>• endDate<br>• minCredit<br>• maxCredit<br>• createdAt<br>• updatedAt</td></tr>
<tr><th colspan="2">Offered Course</th><th colspan="2">User</th></tr>
<tr><th>Client Model</th><th>Database Model</th><td colspan="2">• _id</td></tr>
</table>

| | | |
|---|---|---|
| ● _id<br>● semesterRegistration<br>● academicSemester<br>● academicFaculty<br>● academicDepartment<br>● course<br>● faculty<br>● maxCapacity<br>● remainingCapacity<br>● section<br>● days<br>● startTime<br>● endTime<br>● createdAt<br>● updatedAt | ● semesterRegistration<br>● academicDepartment<br>● course<br>● faculty<br>● maxCapacity<br>● remainingCapacity<br>● section<br>● days<br>● timeSlot | ● id (generated)<br>● password<br>● needsPasswordChange<br>● role<br>● status<br>● createdAt<br>● updatedAt<br>● isDeleted |

| Student | Faculty |
|---|---|
| ● _id<br>● id (generated)<br>● user<br>● name<br>  ○ firstName<br>  ○ middleName<br>  ○ lastName<br>● gender<br>● dateOfBirth<br>● email<br>● contactNo<br>● emergencyContactNo<br>● bloodGroup<br>● presentAddress<br>● permanentAddress<br>● guardian<br>  ○ father<br>  ■ name<br>  ● firstName<br>  ● middleName<br>  ● lastName<br>  ■ occupation<br>  ■ contactNo<br>  ■ email<br>  ○ mother | ● _id<br>● id (generated)<br>● user<br>● designation<br>● name<br>  ○ firstName<br>  ○ middleName<br>  ○ lastName<br>● gender<br>● dateOfBirth<br>● email<br>● contactNo<br>● emergencyContactNo<br>● bloodGroup<br>● presentAddress<br>● permanentAddress<br>● profileImage<br>● academicDepartment<br>● joiningDate<br>● nationality<br>● religion<br>● isDeleted |
| | **Admin** |

| | |
|---|---|
| ■ name<br>● firstName<br>● middleName<br>● lastName<br>■ occupation<br>■ contactNo<br>■ email<br>● localGuardian<br>○ name<br>■ firstName<br>■ middleName<br>■ lastName<br>○ occupation<br>○ contact No<br>○ email<br>○ address<br>○ relationship<br>● profileImage<br>● admissionSemester<br>● nationality<br>● Religion<br>● isDeleted | ● _id<br>● id (generated)<br>● user<br>● designation<br>● name<br>○ firstName<br>○ middleName<br>○ lastName<br>● gender<br>● dateOfBirth<br>● email<br>● contactNo<br>● emergencyContactNo<br>● bloodGroup<br>● presentAddress<br>● permanentAddress<br>● profileImage<br>● managementDepartment<br>● joiningDate<br>● nationality<br>● religion<br>● isDeleted |

## 3. API Endpoints and CRUD Operation Methods:

### a. Academic Faculty:

i. **Create an academic faculty:** academic-faculties/create-academic-faculty **(POST)**
ii. **Get all academic faculties:** academic-faculties **(GET)**
iii. **Get a specific academic faculty:** academic-faculties/:id **(GET)**
iv. **Update a specific academic faculty:** academic-faculties/:id **(PATCH)**

### b. Academic Department:

i. **Create an academic department:** academic-departments/create-academic-department **(POST)**
ii. **Get all academic departments:** academic-departments **(GET)**
iii. **Get a specific academic department:** academic-departments/:id **(GET)**
iv. **Update a specific academic department:** academic-departments/:id **(PATCH)**

### c. Management Department:

i. **Create a management department:** management-departments/create-management-department **(POST)**
ii. **Get all management departments:** management-departments **(GET)**
iii. **Get a specific management department:** management-departments/:id **(GET)**
iv. **Update a specific management department:** management-departments/:id **(PATCH)**

### d. Course:

i. **Create a course:** courses/create-course**(POST)**
ii. **Get all courses:** courses**(GET)**
iii. **Get a specific course:** courses/:id **(GET)**
iv. **Update a specific course:** courses/:id **(PATCH)**
v. **Delete a course:** courses/:id **(DELETE)**
vi. **Get all assigned faculties of a course:** courses/faculties/:id **(GET)**
vii. **Assign faculties to a course:** courses/:id/assign-faculties **(PUT)**
viii. **Remove faculties from a course:** courses/:id/remove-faculties **(DELETE)**

### e. Academic Semester:

i. **Create an academic semester:** academic-semesters/create-academic-semester **(POST)**
ii. **Get all academic semesters:** academic-semesters **(GET)**
iii. **Get a specific academic semester:** academic-semesters/:id **(GET)**
iv. **Update a specific academic semester:** academic-semesters/:id **(PATCH)**

f. **Semester Registration:**

   i. **Create a semester registration:** semester-registrations/create-semester-registration **(POST)**
   ii. **Get all semester registrations:** semester-registrations **(GET)**
   iii. **Get a specific semester registration:** semester-registrations/:id **(GET)**
   iv. **Update a specific semester registration:** semester-registrations/:id **(PATCH)**
   v. **Update all semester registration statuses:** semester-registrations/update/status **(PATCH)**

g. **Offered Course:**

   i. **Create an offered course:** offered-courses/create-offered-course **(POST)**
   ii. **Get all offered courses:** offered-courses **(GET)**
   iii. **Get a specific offered course:** offered-courses/:id **(GET)**
   iv. **Update a specific offered course:** offered-courses/:id **(PATCH)**

h. **User:**

   i. **Create different types of users:**
      1. **Student:** users/create-student **(POST)**
      2. **Faculty:** users/create-faculty **(POST)**
      3. **Admin:** users/create-admin **(POST)**

i. **Student:**

   i. **Get all students:** students **(GET)**
   ii. **Get a specific student:** student/:id **(GET)**
   iii. **Update a specific student:** student/:id **(PATCH)**
   iv. **Delete a specific student:** student/:id **(DELETE)**
   v. **Get students' profile:** students/my-profile **(GET)**

j. **Faculty:**

   i. **Get all faculties:** faculties **(GET)**
   ii. **Get a specific faculty:** faculty/:id **(GET)**
   iii. **Update a specific faculty:** faculty/:id **(PATCH)**
   iv. **Delete a specific faculty:** faculty/:id **(DELETE)**
   v. **Get faculties' profile:** faculties/my-profile **(GET)**

k. **Admin:**

   i. **Get all admins:** admins **(GET)**
   ii. **Get a specific admin:** admin/:id **(GET)**
   iii. **Update a specific admin:** admin/:id **(PATCH)**
   iv. **Delete a specific admin:** admin/:id **(DELETE)**
   v. **Get admins' profile:** admins/my-profile **(GET)**

## I. Auth:

i. **Login:** auth/login
ii. **Register:** auth/register
iii. **Refresh Token:** auth/refresh-token
iv. **Change Password:** auth/change-password
v. **Forgot Password:** auth/forgot-password
vi. **Reset Password:** auth/reset-password